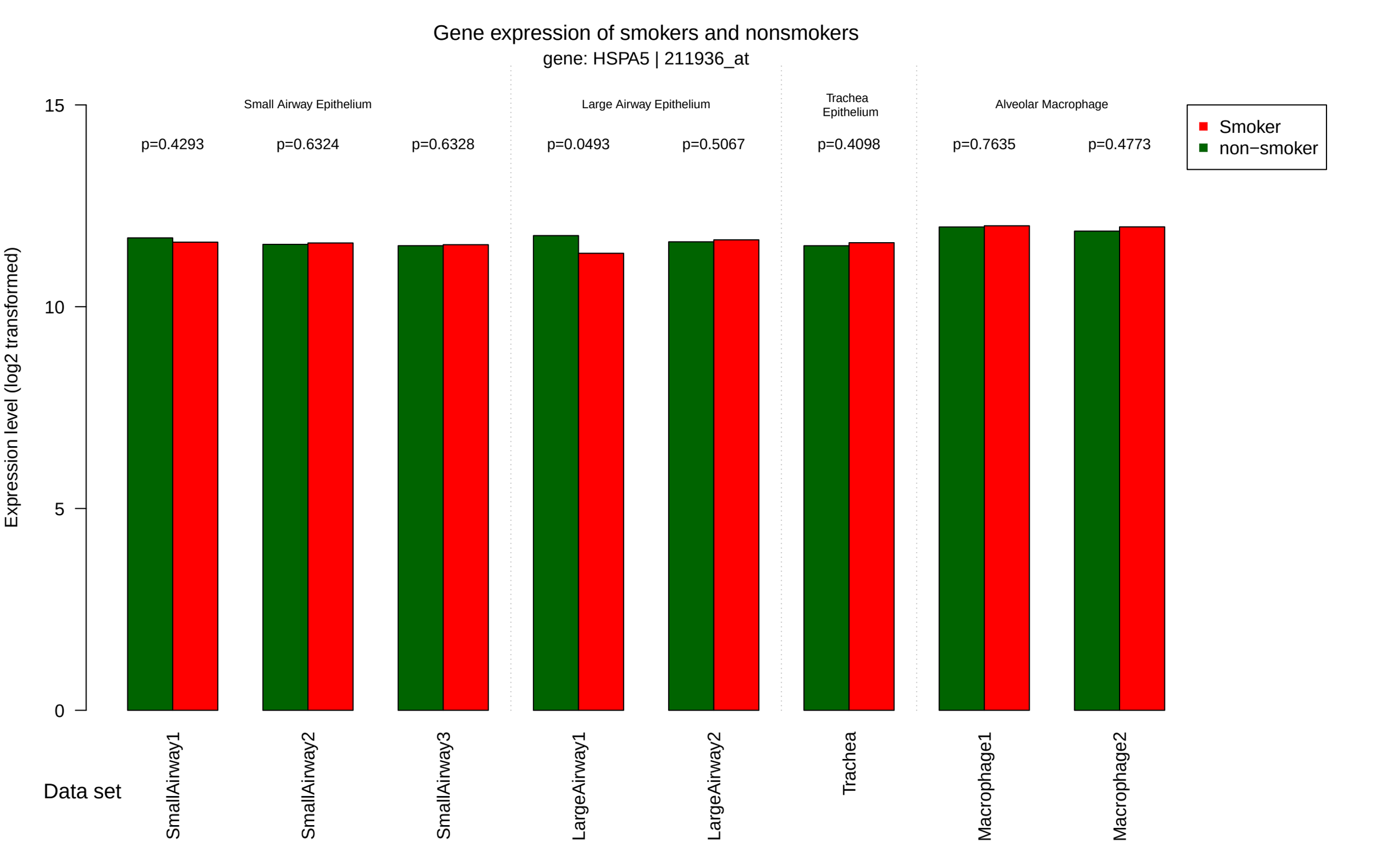
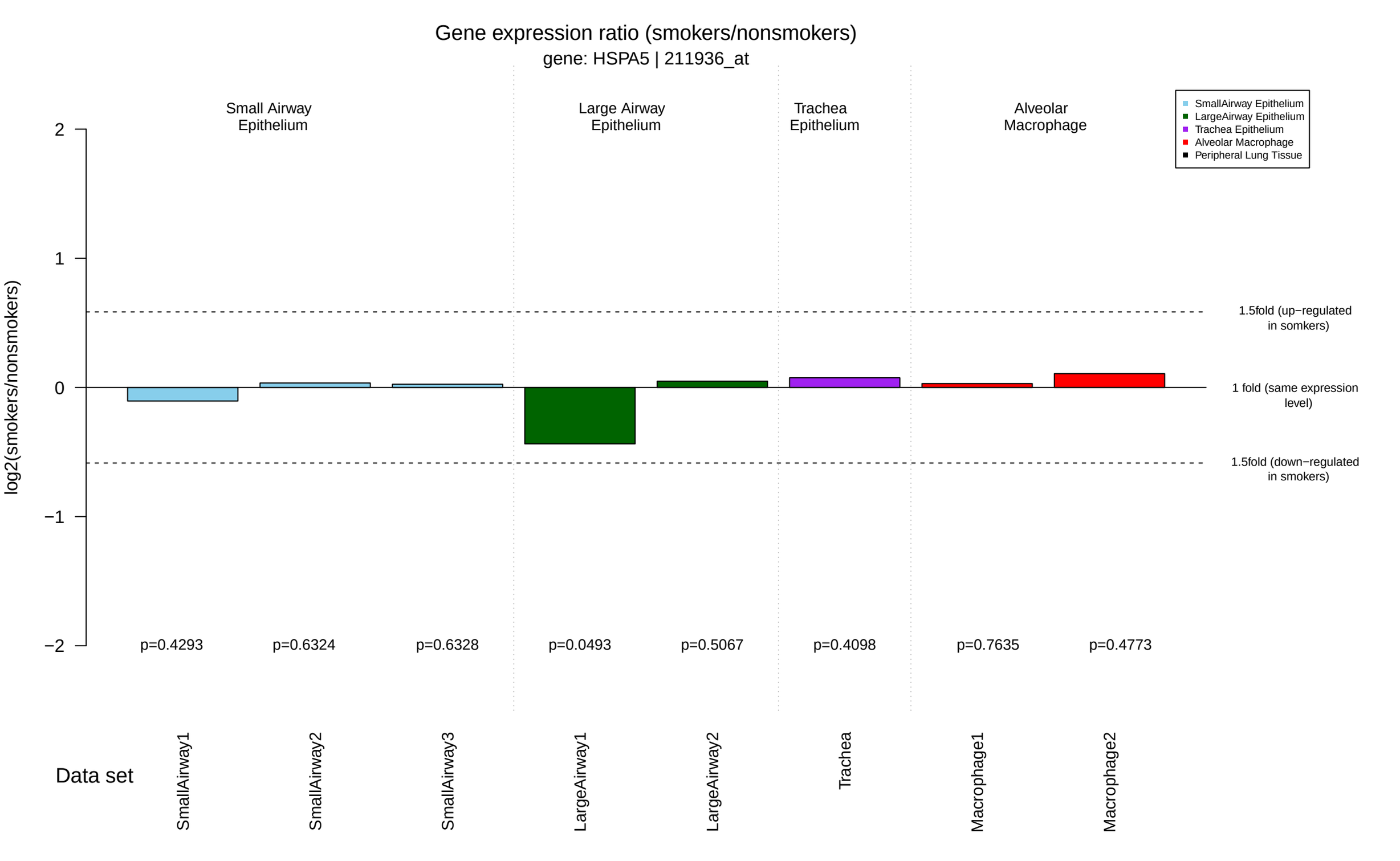
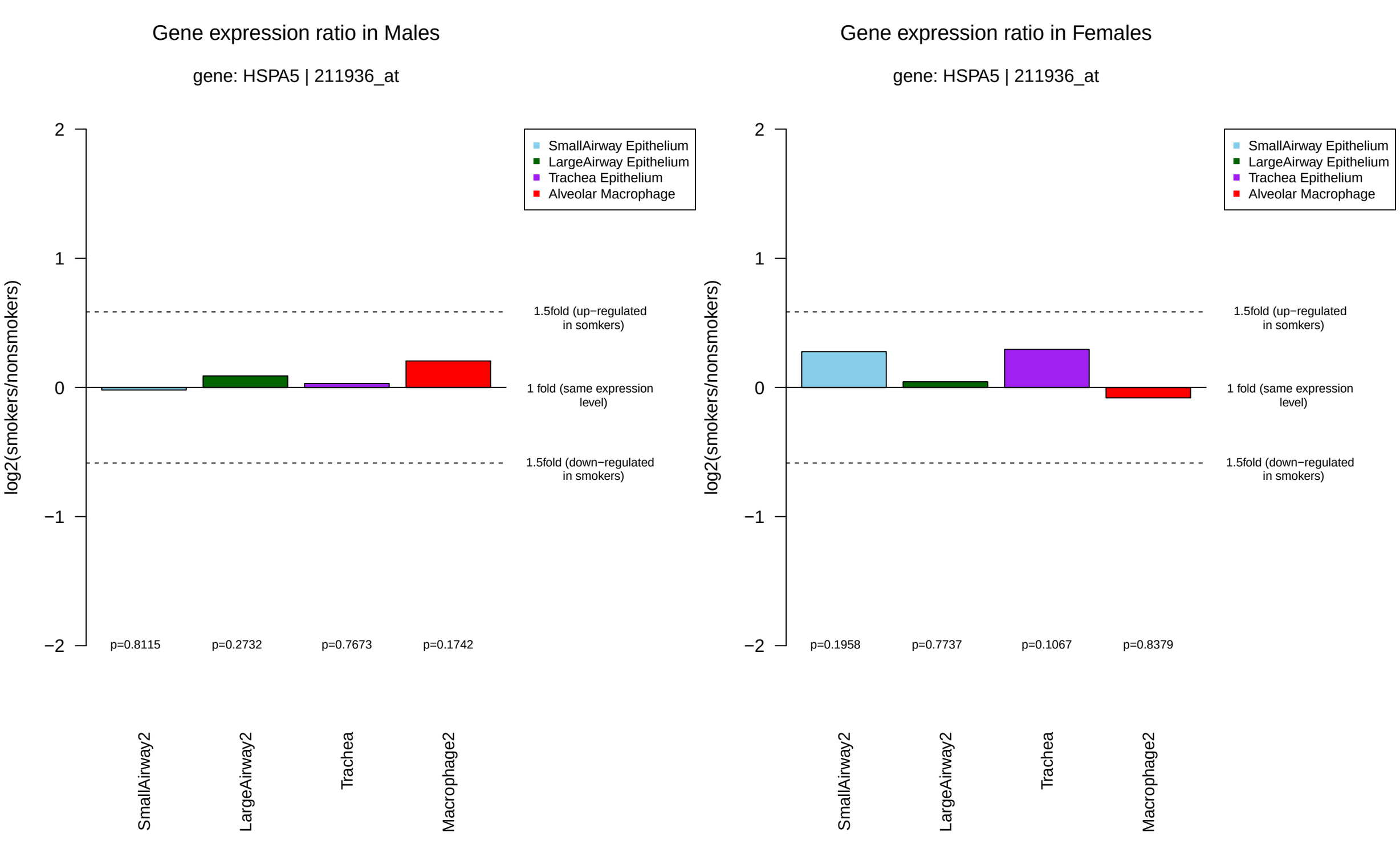
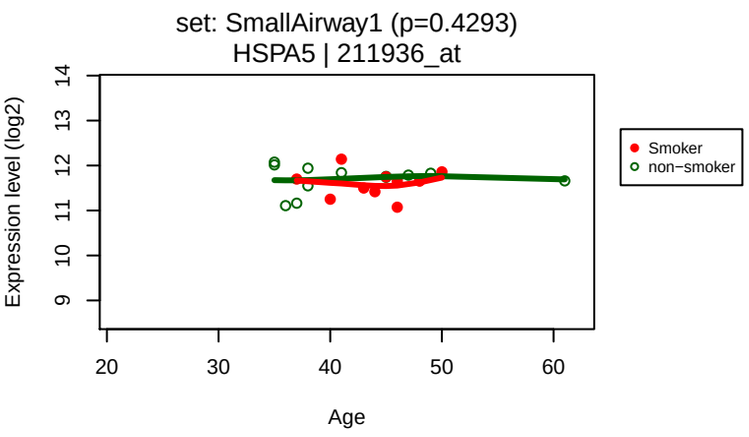
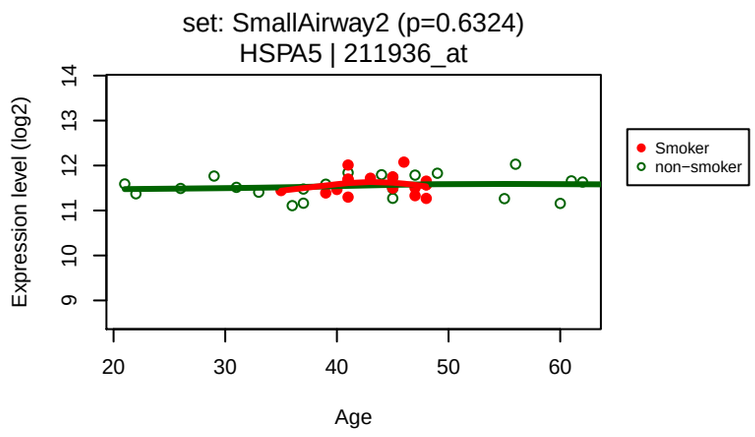
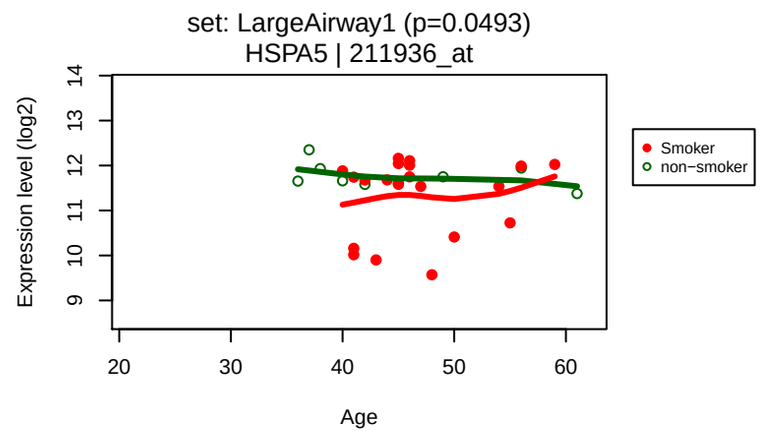

Gene expression of smokers and nonsmokers
gene: HSPA5 | 211936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4293
p=0.6324
p=0.6328
p=0.0493
p=0.5067
p=0.4098
p=0.7635
p=0.4773
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

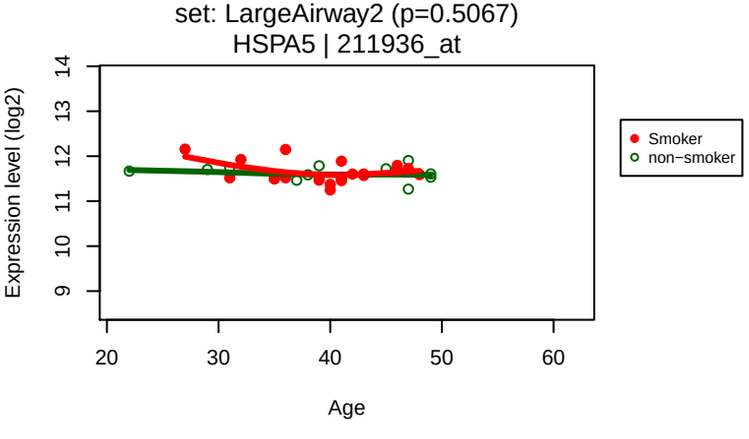

Gene expression ratio (smokers/nonsmokers)
gene: HSPA5 | 211936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4293
p=0.6324
p=0.6328
p=0.0493
p=0.5067
p=0.4098
p=0.7635
p=0.4773
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

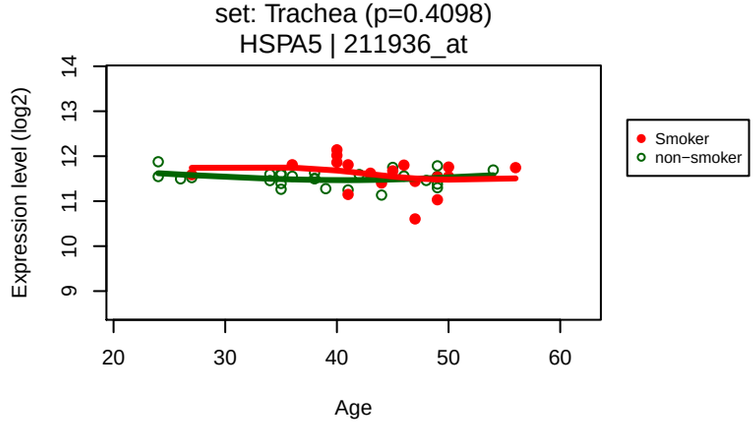

Gene expression ratio in Males
gene: HSPA5 | 211936_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8115
p=0.2732
p=0.7673
p=0.1742
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA5 | 211936_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1958
p=0.7737
p=0.1067
p=0.8379
SmallAirway2
LargeAirway2
Trachea
Macrophage2

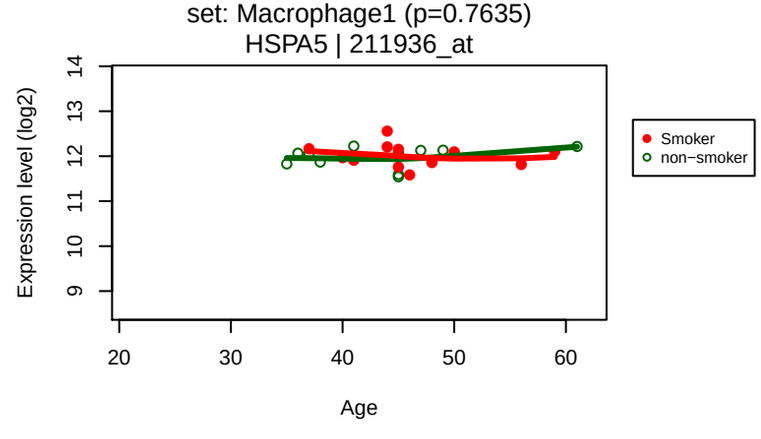

set: SmallAirway1 (p=0.4293)
HSPA5 | 211936_at
Expression level (log2)
Age
Smoker
non−smoker

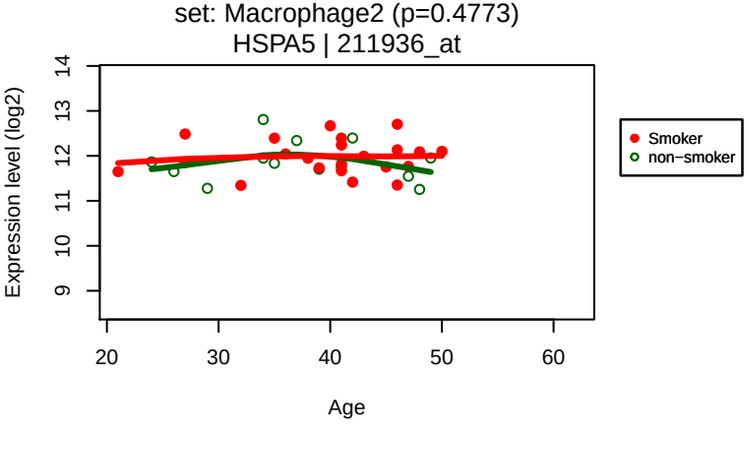

set: SmallAirway2 (p=0.6324)
HSPA5 | 211936_at
Expression level (log2)
Age
Smoker
non−smoker

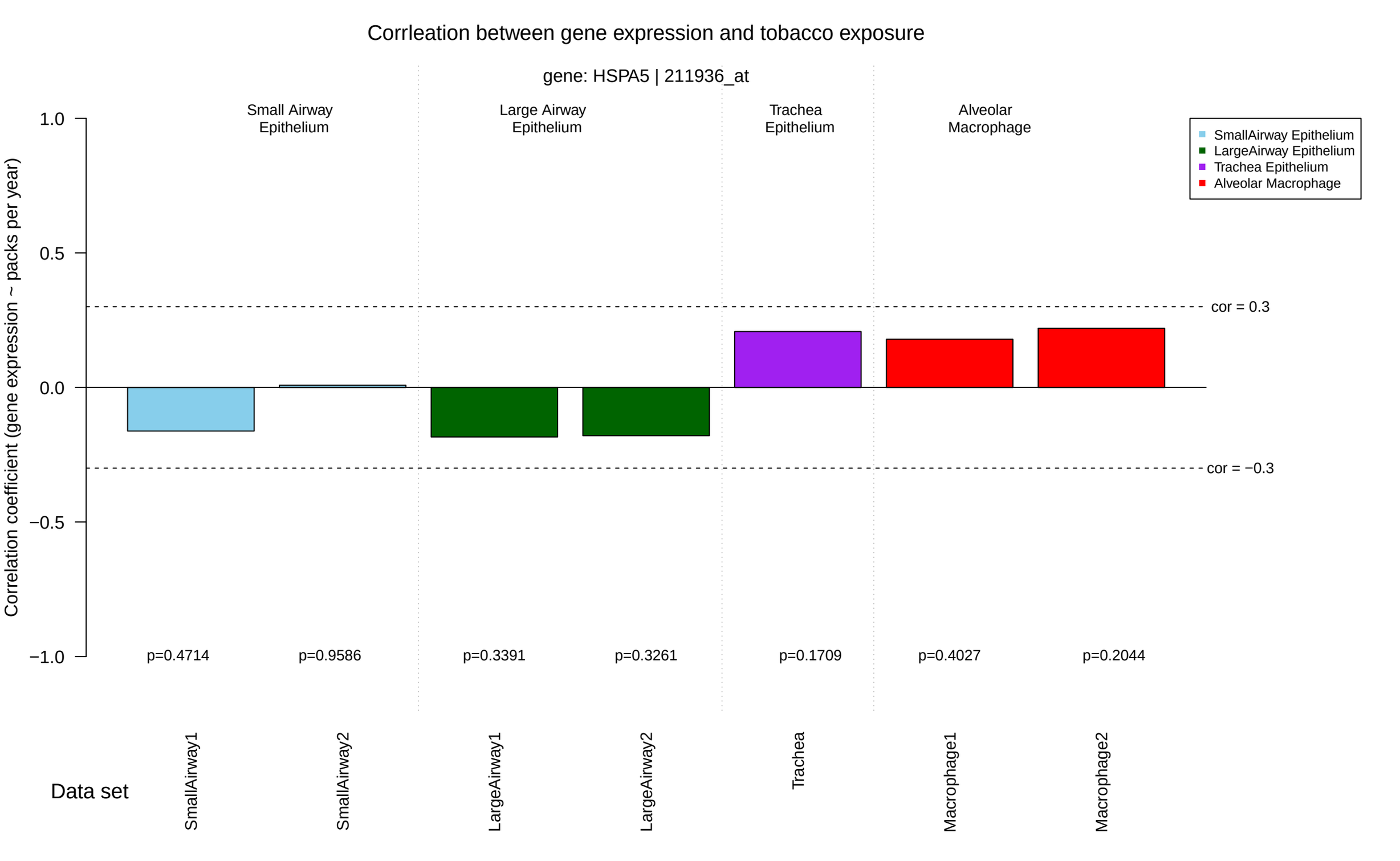

set: LargeAirway1 (p=0.0493)
HSPA5 | 211936_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5067)
HSPA5 | 211936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4098)
HSPA5 | 211936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7635)
HSPA5 | 211936_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4773)
HSPA5 | 211936_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA5 | 211936_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4714
p=0.9586
p=0.3391
p=0.3261
p=0.1709
p=0.4027
p=0.2044
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2